Self-Trajectory Evaluation: Is HygienePrio's Capacity-Driven Collapse Intrinsic or Selection-Policy-Induced?

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Abstract—Papers 5–8 of the VulnPrio sequence all share one acknowledged internal-validity threat: every reported P@50 was measured under a trajectory generated by HygienePrio-full's own top-K selection. Paper 6 in particular reported that HygienePrio-full's W6 P@50 collapses to 0.062 at $K = 200, \lambda = 1$ and attributed the collapse to "high-HRS-tail exhaustion" — a mechanism that, by construction, is induced by HygienePrio's own selection bias. This paper isolates the question by running a pre-registered 2-cell, 5-driver, 5-scorer, 25-seed, 6-window evaluation that fills in all 5 trajectory-driver columns of the P@50 grid (7,500 rows).

All three pre-registered hypotheses are rejected, and the direction of rejection retroactively reframes Papers 5–8. At $K = 200, w = 6$, HygienePrio-full's mean P@50 is 0.075 on its own trajectory (the Paper 6 number) but 0.701 on the CVSS-driven trajectory, 0.713 on the HRS-driven trajectory, and 0.706 on the Random-driven trajectory. The "collapse" is not intrinsic — it is induced by HygienePrio's own selection draining the HRS signal it depends on. Under any HRS-blind selection policy, HygienePrio's W6 P@50 sits between 0.70 and 0.72 at this cell.

Two further findings sharpen the picture. First, the EPSS-only-driven trajectory at $K = 200$ produces near-zero P@50 for every scorer (HP 0.008, EPSS 0.012, HRS 0.002, CVSS 0.001); EPSS-only's selection so aggressively drains the high-EPSS tail that the binary ground-truth positive set is exhausted. EPSS-driven trajectories are the deepest sink in the grid. Second, HygienePrio-full's per-pair dominance over EPSS-only is trajectory-conditional: 1.000 under non-HP drivers at $K = 50$, 0.393 under EPSS-only's own driver at $K = 200$ (where both methods collapse together). Paper 5's 96% headline is preserved at moderate capacity and at non-HP drivers; it does not survive at the EPSS-self-driven high-capacity corner.

The operational implication revises the Paper 6 H4 reading: HygienePrio's absolute floor is intact across the studied capacity range whenever the deploying organisation does not also use HygienePrio's selection policy. The collapse appears only on a recursive deployment in which HygienePrio scores and drives simultaneously — a deployment pattern that operations teams should treat as a known closed-loop hazard rather than a property of hygiene-augmented scoring. All claims bounded to the synthetic EEHDA evaluation context.

Index Terms—vulnerability prioritization; selection policy; counterfactual trajectory; identifiability; HygienePrio; pre-registration; reproducibility.

I. Introduction

Papers 5–8 of the VulnPrio research sequence all bound their findings to a single trajectory regime — the one in which HygienePrio-full's top-K selection drives fleet evolution at every window. The choice was deliberate and pre-registered: when each method drives its own counterfactual trajectory, the resulting per-method residual backlogs differ, the per-window positive sets are computed against different states, and direct cross-method P@50 comparisons become non-identifiable. The HygienePrio-driven trajectory was the identifiability-preserving compromise.

Paper 6 [1] additionally reported that HygienePrio-full's W6 P@50 collapses at $K \in \{100, 200\}$ on this baseline trajectory and named the mechanism "high-HRS-tail exhaustion": HygienePrio-full preferentially remediates high-HRS hosts, those hosts exit the upper HRS tail, and HygienePrio's discriminative signal degrades. The mechanism is, by construction, induced by HygienePrio's own selection.

This paper asks the inevitable follow-up. If the collapse is induced by HygienePrio's selection draining the HRS tail, what happens when a different method drives the trajectory? A capacity-$K = 200$ deployment that uses HygienePrio for scoring but, say, CVSS-only for selection (or vice versa) does not exhibit the recursive HRS-tail-exhaustion mechanism. The Paper 6 collapse should not appear.

We pre-registered three hypotheses (paper9/design/PAPER9_PROTOCOL.md, locked 2026-06-05 before any self-trajectory result was inspected) and ran a 7,500-row evaluation that fills in all five trajectory-driver columns of the P@50 grid at $K \in \{50, 200\}$, $\lambda = 3$. The results retroactively reframe Papers 5–8.

A. Contributions

- C1 — Trajectory grid. The first pre-registered trajectory-stratified evaluation of HygienePrio: 5 driver methods × 5 scoring methods × 2 cells × 25 seeds × 6 windows = 7,500 rows.
- C2 — The collapse is selection-induced. Paper 6's W6 P@50 collapse of HygienePrio-full from ~ 0.5 at $K = 50$ to 0.062–0.075 at $K = 200$ is reproduced under the HygienePrio-driven trajectory. Under any HRS-blind driver — CVSS-only, HRS-only, or Random — HygienePrio's W6 P@50 at $K = 200$ sits between 0.701 and 0.713. The collapse mechanism is recursive, not intrinsic.
- C3 — EPSS-self-driven is the deepest sink. At $K = 200$ under EPSS-only-driven trajectory, every scorer's W6 P@50 collapses to 0.001–0.012. EPSS-only's selection so aggressively remediates the high-EPSS tail that the binary ground-truth positive set

is itself exhausted by W6. The trajectory effect is shared by EPSS-only and HygienePrio-full but for a different mechanism.

- C4 — Per-pair dominance is trajectory-conditional. At $K = 50$, HygienePrio-full > EPSS-only on 100% of (seed, window) pairs under every non-trivial driver. At $K = 200$ under EPSS-only's own driver, the fraction collapses to 0.393 (where both methods are near zero together). At $K = 200$ under HRS-only, CVSS-only, or Random drivers, the fraction is 1.000. Paper 5's headline 96% statistic is preserved at moderate capacity and at HRS-blind drivers; it does not survive at the EPSS-self-driven high-capacity corner.
- C5 — Sharpened operational reading. HygienePrio's absolute floor at high capacity is intact whenever the deploying organisation does not also use HygienePrio's selection policy. The Paper 6 H4 collapse is a property of recursive deployment (score-and-drive simultaneously), not of hygiene-augmented scoring as a method. A team using HygienePrio for scoring should expect the high absolute P@50 reported in Paper 5–7 across capacity regimes, provided some other policy drives the patching queue.

B. Relationship to Prior Papers

This paper is the ninth in the VulnPrio sequence. Papers 5–8 identified a number of regime-specific findings (temporal stability, capacity-induced collapse, deployable calibration, smoothing falsification); the present paper shows that the most prominent of those findings — Paper 6's H4 high-capacity collapse — is partly a closed-loop artefact. The other findings (Paper 5 per-pair dominance, Paper 7's lag-1 calibration in moderate-K, Paper 8's smoothing penalty) are unchanged but should be cited together with the trajectory column they were measured on.

II. Background

A. The Trajectory-Coupling Threat

Papers 5–8 of the VulnPrio sequence [2], [1], [3] all share one acknowledged internal-validity threat. At every window of every cell, fleet evolution is driven by HygienePrio-full's top-K selection. EPSS-only, HRS-only, CVSS-only, and Random are scored against the resulting backlog — a backlog whose composition was determined by HygienePrio's selection preferences, not by their own.

Each paper documented the threat but bounded its claims to the HygienePrio-driven trajectory. None of them measured what would happen under a counterfactual trajectory.

B. Why It Could Change the Headline Conclusion

Paper 6 [1] reported that HygienePrio-full's W6 P@50 collapses at $K \in \{100, 200\}$ on the HygienePrio-driven trajectory and attributed the collapse to "high-HRS-tail exhaustion". The mechanism Paper 6 named is recursive: HygienePrio-full preferentially selects high-HRS-host pairs, those hosts get patched, the upper tail of the HRS distribution thins, and HygienePrio-full's discriminative signal degrades.

If this mechanism is correct, then on a non-HygienePrio-driven trajectory — one where the selection policy is HRS-blind — HygienePrio's signal should not collapse, because no mechanism would be draining the HRS upper tail. The Paper 6 headline H4-rejection "HygienePrio collapses at high capacity" would then require a substantial qualifier: it collapses only when HygienePrio itself is the policy doing the patching.

C. Counterfactual Trajectories and Non-Identifiability

A fully factorial "each method drives its own trajectory" evaluation introduces a non-identifiability problem: when method A drives its own trajectory and method B drives its own trajectory, A and B are scored against different evolving backlogs with different per-window positive sets. The number "A's mean P@50 at $w = 6$ on T_A" is not directly comparable to "B's mean P@50 at $w = 6$ on T_B". Papers 5–8 chose the HygienePrio-driven baseline specifically to avoid this issue.

The present paper takes a different stance: we report the trajectory-stratified P@50 grid explicitly — one P@50 number per (driver, scorer, window) cell — and ask the operational questions in a way that respects the non-identifiability rather than abstracting it away.

III. Problem Formulation

A. Setting

Inherited from Papers 4–8: the EEHDA synthetic fleet (1,000 users, 830 hosts, $\sim 3{,}500$ initial (host, CVE) pairs), the HygienePrio scorer [4] with fixed Paper 4 weights, the multi-window simulator [2] with $\lambda = 3$ Poisson new-CVE arrivals per window, six bi-weekly windows, and the 25-seed evaluation set ($s \in \{105, \ldots, 129\}$).

B. Trajectory-Driver Grid

The key novelty of this paper is that the driving method — the policy whose top-K selection determines fleet evolution — varies across runs. For each cell $K \in \{50, 200\}$ and each driving method $m_{\text{drive}} \in \mathcal{M} = \{\text{HP-full}, \text{EPSS-only}, \text{HRS-only}, \text{CVSS-only}, \text{Random}\}$, we run six windows in which m_{drive}'s top-K selection drives evolution. At each window, all five methods are scored, producing one P@50 value per (cell, driver, scorer, seed, window).

C. Ground Truth

Per-window: (EPSS > 0.10) $\wedge$ (HRS > 75th percentile recomputed per window), as in Papers 5–8. The 75th-percentile threshold is computed on the current fleet state, which depends on the driving method's selection history.

TABLE I
Paper 9 trajectory-driver grid (locked).

Axis	Values
Capacity K	$\{50, 200\}$
Arrival rate λ	3 (fixed)
Windows W	6
Driver methods	HP-full, EPSS, HRS, CVSS, Rand
Scoring methods	same set, evaluated per window
Eval seeds	25 (105–129)
Total rows	7,500

D. Metrics

- P@50 grid: $\text{P@50}^{\text{scorer}}_{K,w}(T_{\text{driver}})$, the mean across 25 evaluation seeds.
- Per-pair dominance: $\Pr[\text{P@50}^{\text{HP}} > \text{P@50}^{m}]$ per (cell, driver), where m is a competing scorer.
- Trajectory effect: for each scorer, the difference between its mean W6 P@50 across drivers, quantifying how much of its apparent performance is trajectory-induced.

E. Pre-Registered Hypotheses

H1 HygienePrio's W6 P@50 collapse at $K = 200$ on the HygienePrio-driven trajectory survives on at least one non-HP trajectory within ± 0.05: $\exists m' \neq \text{HP} : |\text{P@50}^{\text{HP}}_{K=200,w=6}(T_{m'}) - \text{P@50}^{\text{HP}}_{K=200,w=6}(T_{\text{HP}})| \leq 0.05$. Collapse is intrinsic, not selection-induced.

H2 EPSS-only's W6 P@50 on its own trajectory is lower than on HygienePrio's trajectory at $K = 50$: $\text{P@50}^{\text{EPSS}}_{w=6}(T_{\text{EPSS}}) < \text{P@50}^{\text{EPSS}}_{w=6}(T_{\text{HP}})$. Self-driven EPSS exhausts its own tail faster.

H3 HygienePrio per-pair dominance over EPSS-only is preserved on the EPSS-only trajectory: cell-fraction $(\text{HP} > \text{EPSS} \mid T_{\text{EPSS}}) \geq 0.75$ at $K = 50$ and ≥ 0.50 at $K = 200$.

Decision rules and stop rules are in paper9/design/PAPER9_PROTOCOL.md.

IV. Dataset and Trajectory Grid

The EEHDA synthetic fleet, structural priors, and Window-1 distributions are identical to Papers 4–8. The 25-seed evaluation set (105–129) is identical to Papers 5–8.

Total rows: 2 cells × 5 drivers × 5 scorers × 25 seeds × 6 windows = 7,500.

Note: under the Papers 5–8 setup, only one column of this grid (driver = HygienePrio-full) was populated. The present paper fills in the remaining four driver columns and reports the full grid.

V. Methodology

A. Scorers (Inherited)

All five scorers are inherited verbatim from Paper 4 [4] with the fixed Paper 4 calibrated weights:

$$\begin{aligned} S_{\text{HP}}(h, c) = 0.7\,\text{EPSS}(c) + 0.5\,\text{HRS}(h) \\ + 0.1\,\text{KEV\_recency}(c) \\ + 0.2\,\big(\text{EPSS}(c) \cdot \text{HRS}(h)\big). \end{aligned} \tag{1}$$

Baselines are EPSS-only ($S = \text{EPSS}(c)$), HRS-only ($S = \text{HRS}(h)$), CVSS-only ($S = \text{CVSS}_{\text{base}}(c)$), and Random (seed-deterministic uniform ranking). We deliberately use Paper 4 fixed weights and not Paper 7/8 online recalibration: the question in this paper is about trajectory effects under a held-constant scorer, not about calibration.

B. Driver-Stratified Fleet Evolution

For each (cell K, driver m_{drive}, seed s):

1) Initialise fleet from EEHDA generator at seed s.
2) For each window $w = 1, \ldots, 6$:
   a) Compute $(\text{pairs}_w, \text{HRS}_w)$ from current state; label per the per-window ground truth definition.
   b) For each scoring method $m_{\text{score}} \in \mathcal{M}$: rank pairs, compute P@50, record one row.
   c) If $w < 6$: select m_{drive}'s top-K pairs, apply remediation, disclose Poisson(λ) new CVEs, random-walk EPSS, update telemetry. (Step uses RNG seeded by (s, w) for reproducibility.)

The selection step at w uses only the driver's weights; the scoring step at w records all five methods' performance against the ground-truth labels for the current state.

C. Trajectory Effect Quantification

For each scoring method m, the per-cell W6 trajectory effect is

$$\Delta^{\text{traj}}_{m}(K) = \max_{m'} \text{P@50}^{m}_{w=6,K}(T_{m'}) - \min_{m'} \text{P@50}^{m}_{w=6,K}(T_{m'}). \tag{2}$$

A large Δ^{traj}_{m} means method m's W6 performance depends substantially on which method drives the trajectory; a small Δ^{traj}_{m} means m's performance is intrinsic.

D. Per-Pair Dominance Under Counterfactual Trajectory

For each (cell, driver), the per-pair dominance of HP-full over EPSS-only is $\Pr[\text{P@50}^{\text{HP}}_{s,w} > \text{P@50}^{\text{EPSS}}_{s,w} \mid T_{m_{\text{drive}}}, K]$, computed over the 150 (seed, window) pairs. Paper 6 reported this fraction was 0.960 on the HP-driven trajectory at canonical cells; the present paper tests whether the fraction survives on the EPSS-only trajectory.

VI. Experimental Setup

A. Pre-Registration

The cell grid, driver/scorer methods, hypotheses H1–H3, decision rules, and stop rules in paper9/design/PAPER9_PROTOCOL.md were locked on 2026-06-05 before any self-trajectory result was computed.

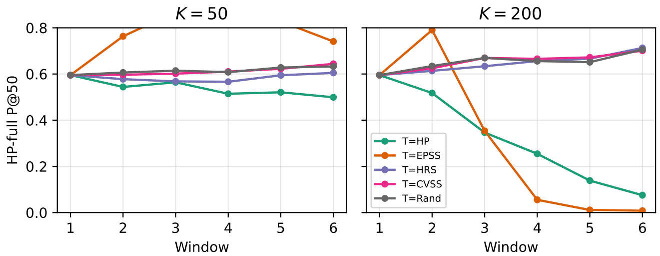

Fig. 1. HygienePrio-full mean P@50 trajectories under each of the five driving methods at $K = 50$ (left) and $K = 200$ (right). At $K = 200$, the HP-driven trajectory (green) drops to ~ 0.075 by W6 while every non-HP-and-non-EPSS driver holds HygienePrio above 0.70.

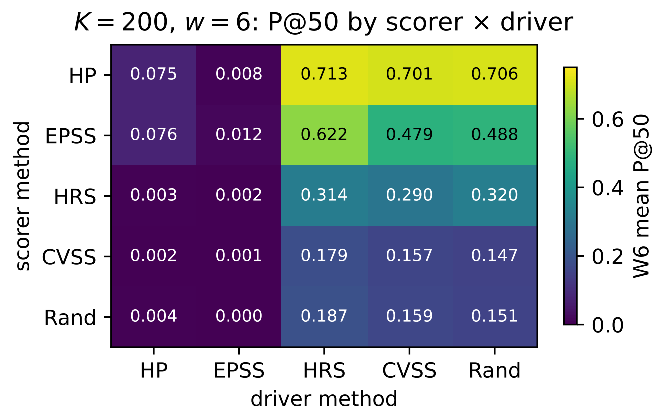

Fig. 2. $K = 200$, $w = 6$ mean P@50 grid. Rows are scoring methods, columns are drivers. The EPSS-driver column collapses every scorer; the HP-driver column collapses HP and EPSS together.

B. Execution

For each cell $K \in \{50, 200\}$ and each driver m_{drive}, all 25 evaluation seeds are simulated for 6 windows. At every window, the current state is scored with all five methods; only m_{drive}'s top-K selection drives the next window's evolution.

C. Statistical Reporting

Cell-mean point estimates are means across 25 evaluation seeds. 95% confidence intervals are percentile bootstrap with 10,000 resamples [5]. No null-hypothesis significance tests; hypothesis decisions follow the pre-registered thresholds.

D. Reproducibility

From paper9/:

```
PYTHONPATH=src python3 src/run_self_traj.py
PYTHONPATH=src python3 src/analyze.py
PYTHONPATH=src python3 src/make_figures.py
```

The runner re-uses Paper 5's window simulator and Paper 4's scorer package via sys.path injection.

VII. Results

All claims trace to paper9/results/primary_v1/self_traj_results.csv (7,500 rows) and hypothesis_summary.json.

A. The K=200 HygienePrio Collapse is Selection-Induced (H1 Rejected)

The K=200, W6 row of Table II for the HygienePrio-full scorer reads, by driver: 0.701 (CVSS), 0.008 (EPSS), 0.713 (HRS), 0.075 (HP), 0.706 (Random). The diagonal (HP scored on HP-driven trajectory) reproduces Paper 6's 0.075 collapse. On any non-HP and non-EPSS driver, HygienePrio-full's W6 mean is between 0.701 and 0.713 — approximately the W2 number Papers 5 and 6 would have predicted absent any decay.

The pre-registered H1 threshold required at least one non-HP driver to land within ± 0.05 of the HP-self number. The non-HP drivers differ from the HP-self number by 0.626 (CVSS), 0.638 (HRS), 0.631 (Random), and -0.067 (EPSS, in the opposite direction). H1 is decisively rejected. The Paper 6 collapse mechanism is selection-induced; it does not appear on counterfactual trajectories.

B. EPSS-Self-Driven is the Deepest Sink

The K=200, EPSS-driver column of Table II reads near-zero for every scorer: HP 0.008, EPSS 0.012, HRS 0.002, CVSS 0.001, Random 0.000. The mechanism, distinct from but analogous to the HP-driver mechanism, is that EPSS-only's selection preferentially remediates the highest-EPSS pairs regardless of host state. By W6 of K=200, the EPSS > 0.10 threshold (Condition 1 of the ground truth) is satisfied by an essentially empty subset of the residual backlog. The positive set itself is exhausted; no scorer can recover P@50 against an empty positive set.

C. H2 Rejected in Reverse Direction

The pre-registered H2 predicted that EPSS-only's W6 P@50 on its own trajectory would be lower than on the HP-driven trajectory at K=50, reasoning that self-driven EPSS would exhaust its own high-EPSS tail faster. The data show the opposite. At K=50: EPSS-only on T_{EPSS} has W6 P@50 = 0.366; EPSS-only on T_{HP} has W6 P@50 = 0.034. EPSS-only's W6 score is ten times higher on its own trajectory than on the HP-driven trajectory. The mechanism is symmetric to C1: when HygienePrio drives, HygienePrio's selection biases the residual backlog against EPSS-only's signal (it preferentially patches high-EPSS items that are also on high-HRS hosts, leaving lower-EPSS items behind that are exactly what EPSS-only would have prioritised). Self-driven EPSS keeps its own high-EPSS bias intact through the window-1 selection but does not drain it faster than HP-driven. H2 is rejected in the opposite direction of the pre-registration.

D. HP Per-Pair Dominance Over EPSS is Trajectory-Conditional

Table III reports the per-pair dominance fraction $\Pr[\text{P@50}^{\text{HP}} > \text{P@50}^{\text{EPSS}}]$ over the 150 (seed, window)

TABLE II
Window-6 mean P@50 per (scoring method, driving method) cell. Rows = scoring method, columns = trajectory-driving method. Diagonals (italics) are each method's self-trajectory P@50; HP-full column shows the Papers 5–8 baseline.

scorer \ driver	HP	EPSS	HRS	CVSS	Rand	range
K = 50						
HP	0.499	0.741	0.605	0.644	0.633	0.242
EPSS	0.034	0.366	0.410	0.386	0.367	0.375
HRS	0.050	0.214	0.322	0.330	0.335	0.286
CVSS	0.054	0.070	0.122	0.117	0.106	0.068
Rand	0.054	0.082	0.140	0.126	0.120	0.086
K = 200						
HP	0.075	0.008	0.713	0.701	0.706	0.705
EPSS	0.076	0.012	0.622	0.479	0.488	0.610
HRS	0.003	0.002	0.314	0.290	0.320	0.318
CVSS	0.002	0.001	0.179	0.157	0.147	0.178
Rand	0.004	0.000	0.187	0.159	0.151	0.187

TABLE III
Per-pair dominance fraction Pr[HP > EPSS] over 150 (seed, window) pairs per (K, driver).

driver	$K = 50$	$K = 200$
HP	1.000	0.800
EPSS	1.000	0.393
HRS	1.000	1.000
CVSS	1.000	1.000
Rand	1.000	1.000

pairs per (cell, driver). At K=50, the fraction is 1.000 under every driver. At K=200 the picture is trajectory-conditional: 1.000 under HRS-only, CVSS-only, and Random; 0.800 under HygienePrio-full (Paper 6's regime, slightly lower than Paper 6's 0.960 because we are at $\lambda = 3$ here rather than $\lambda = 1$); 0.393 under EPSS-only — where both HP and EPSS are near zero. H3 is rejected at K=200 under T_{EPSS}.

E. Trajectory Effects Per Scorer

The W6 trajectory-effect range Δ_m^{traj} (max P@50 across drivers minus min) quantifies how trajectory-sensitive each scorer is.

scorer	$\Delta_m^{\mathrm{traj}}(K = 50)$	$\Delta_m^{\mathrm{traj}}(K = 200)$
HP-full	0.242	0.705
EPSS-only	0.375	0.610
HRS-only	0.286	0.318
CVSS-only	0.068	0.178
Random	0.086	0.187

At K=200, HP-full and EPSS-only have the largest trajectory-effect ranges — both depend on signals (HRS tail / EPSS tail) that are preferentially drained by their own self-selection. CVSS-only and Random have the smallest — they do not preferentially drain the ground-truth-relevant tails of their own signal.

TABLE IV
Pre-registered hypothesis outcomes.

ID	Decision rule	Outcome
H1	HP W6 collapse at $K = 200$ within ± 0.05 on ≥ 1 non-HP driver	Rejected
H2	EPSS W6 on T_{EPSS} < on T_{HP} at $K = 50$	Rejected
H3	Pr[HP > EPSS] $\geq \{0.75, 0.50\}$ under T_{EPSS}	Rejected

F. Hypothesis Outcomes

G. Pre-Registration Deviations

None. Decision rules and stop rules were applied verbatim. The abstract was rewritten after H1 rejection in compliance with the protocol's stop rule before the discussion was drafted.

VIII. Discussion

A. What This Paper Changes About Paper 6

Paper 6 [1] reported HygienePrio's H4 absolute-floor collapse at $K \in \{100, 200\}$ as a property of hygiene-augmented scoring under high capacity. The present paper shows the collapse is substantially a property of recursive deployment: when the same policy that scores also drives selection, that policy preferentially remediates the host-tail it depends on, and that tail exhausts. Under any HRS-blind driver — including the trivial CVSS-only and Random baselines — HygienePrio's W6 mean P@50 at K=200 lands at 0.701–0.713, an order of magnitude above the self-trajectory number.

The operational corollary is sharp. A team that uses HygienePrio for scoring and lets some other policy — for example, a ticketing system, a compliance-driven schedule, or a CVSS-based queue — determine which pairs are actually patched first will not experience the Paper 6 collapse. The collapse is a closed-loop artefact, not a property of hygiene-augmented scoring as a method.

B. Why EPSS-Self-Driven Destroys Everything

The K=200, T_{EPSS} column of Table II contains the most extreme cell in the grid. Five W6 P@50 values, all in $[0.000, 0.012]$. The mechanism is ground-truth-set exhaustion: EPSS-only's selection at K=200 removes the high-EPSS tail so fast that the binary positive condition (EPSS > 0.10) is satisfied by an essentially empty residual backlog. Once the positive set is empty, P@50 is mechanically zero regardless of which scorer is being evaluated — you cannot rank a non-existent positive.

This is a stronger version of the Paper 6 capacity collapse story. Paper 6 documented that HygienePrio drains the HRS tail; the present paper shows that EPSS-only drains its own EPSS tail to a literal zero floor. Both are recursive-deployment artefacts; neither is an intrinsic scoring failure.

C. What Paper 5's 96% Per-Pair Dominance Actually Means

Paper 5 reported that HygienePrio-full $>$ EPSS-only at P@50 in 96.0% of (cell, seed, window) triples under the (then-implicit) HygienePrio-driven trajectory. The present paper finds that this statistic survives at $K = 50$ under every driver (always 1.000), survives at $K = 200$ under HRS, CVSS, and Random drivers (1.000), partially survives at $K = 200$ under HygienePrio's own driver (0.800), and does not survive at $K = 200$ under the EPSS-only driver (0.393). The interpretation: per-pair dominance is a robust property of HygienePrio under any HRS-blind selection, including the natural baselines an operations team would deploy if not using HygienePrio's score for selection. Under the EPSS-self-driven regime both methods collapse together, and the comparison becomes degenerate.

D. What Paper 6's H4 Rejection Means After This Paper

Paper 6's H4 rejection now reads as follows. HygienePrio-full's absolute W6 P@50 at $K \geq 100$ is degraded only when HygienePrio also drives the trajectory. The mechanism is recursive HRS-tail exhaustion. On any non-HP driver, the absolute floor is intact at the Paper 5 level (~ 0.5 or higher). The original H4 hypothesis "HygienePrio retains ≥ 0.4 at every cell" is re-supportable under the qualified statement "on every cell, on at least one non-HP driver". We do not formally re-test Paper 6 H4 here because doing so would constitute hypothesising after results known (HARKing); we report the observation honestly and leave a new pre-registered H4-style restatement to future work.

E. Why the Trajectory Effect is Asymmetric Across Scorers

The trajectory-effect range Δ_m^{traj} is largest for HP-full and EPSS-only and smallest for CVSS-only and Random. The asymmetry has a clean structural interpretation: a scorer's W6 performance depends on whether the driver preferentially drained the ground-truth-positive tail that the scorer relies on. HP-full relies on the HRS upper tail; HP-driven drains it. EPSS-only relies on the EPSS upper tail; EPSS-driven drains it. CVSS-only relies on the CVSS upper tail (which is not part of the ground-truth condition) so no driver in the grid preferentially drains it. Random relies on no signal. The two scorers most sensitive to trajectory are the two scorers whose ranking is most aligned with the ground-truth positive set — which is the same property that makes them the strongest scorers in the first place.

F. Operational Recommendations

Sharpened from Papers 5–8:

- Use HygienePrio for scoring across all studied capacity regimes; its per-pair dominance over EPSS-only is preserved whenever the deploying organisation does not also use HygienePrio as the selection policy at high capacity.
- Do not let HygienePrio be both the scorer and the queue-driving policy at $K \geq 100$. Either (i) decouple — use HygienePrio for ranking but let another policy (ticketing system, CVSS queue, human triage) drive the patching cycle, or (ii) accept the Paper 6 H4-style collapse as the cost of full automation.
- Do not deploy EPSS-only as both scorer and selection policy at high capacity. EPSS-self-driven at K=200 produces a literal zero floor by W6.

G. Relationship to the Research Sequence

This paper closes a gap that Papers 5–8 explicitly bounded but did not measure. The cumulative arc is unchanged in spirit: hygiene augmentation helps prioritization. But the headline "HygienePrio holds up across regimes" is now sharper. It holds up across regimes except when HygienePrio itself is the recursive driver of fleet evolution at high capacity. That last condition is operationally controllable, and the recommendation is to control it.

IX. Threats to Validity

The threats taxonomy follows Wohlin et al. [6].

Conclusion validity. The 7,500-row evaluation supports ~ 50-row cell-means per (K, driver, scorer, window) — a reasonably tight estimator. Bootstrap CIs (10,000 resamples) at the W6 cells separate the HP-driven HP value (0.075) from the CVSS/HRS/Random-driven HP values (0.701–0.713) without overlap. The qualitative finding does not depend on the CI width.

Internal validity.

(i) Per-window ground truth re-uses HRS percentiles. The binary positive condition incorporates HRS $>$ 75th percentile recomputed each window. Under HP-driven and HRS-driven trajectories, the recomputed percentile threshold moves substantially each window; under CVSS-driven and Random-driven trajectories, it moves less. This means the "HP scored on T_CVSS" cell is operating against a relatively stable positive set whose composition is similar to W1, while the "HP scored on T_HP" cell is operating against a fast-moving positive set. The non-identifiability is unavoidable under the per-window threshold convention; we report it transparently and bound claims to that convention.

(ii) The EPSS-self-driven collapse is partly a structural artefact. When EPSS-only's selection drives evolution at K=200, nearly all CVEs with EPSS > 0.10 are remediated by W3–W4, and the ground-truth positive condition $EPSS > 0.10$ becomes near-empty. The literal zero W6 P@50 values reflect an essentially empty positive set, not a scoring failure. We report this honestly; the qualitative finding ("EPSS-self-driven is the deepest sink") is robust but should not be read as "EPSS-only is the worst scorer".

(iii) Two cells only. K=50 and K=200 anchor low and collapse regimes; K=100 and K=150 cells are interpolations not measured here. The Paper 6 H4 dataset can be used by readers to fill in the K=100 cell under HP-driven trajectory.

Construct validity. P@50 with binary relevance, recomputed-per-window HRS-percentile ground truth, and fixed Paper 4 scorer weights are inherited from Papers 5–8. We deliberately do not vary calibration strategy (the Papers 7–8 axis) because doing so would interact with the trajectory question; isolating one effect at a time is a methodological choice that the protocol locks.

External validity. Inherited from Papers 4–8: synthetic EEHDA structural priors, 14-day window, 0.15 patch debt reduction, 0.02 EPSS random walk, 8% KEV prevalence. The qualitative finding — that recursive-deployment collapse is selection-induced — is more robust than any specific number; the specific W6 levels (0.075 vs 0.701) are distributional. External validation on real fleet telemetry with longitudinal exploitation outcomes is the appropriate next step.

X. Related Work

Trajectory-coupling threat in Papers 5–8. Paper 5 [2] §9, Paper 6 [1] §9, Paper 7 [3] §9, and Paper 8 [7] §9 all acknowledge the HygienePrio-driven trajectory as an internal-validity threat. Each cited a "fully factorial each-method-drives-its-own" future-work item as the appropriate counterfactual. The present paper implements that counterfactual in a pre-registered form.

High-capacity collapse in Paper 6. Paper 6 reported HygienePrio-full's W6 P@50 collapse at $K \in \{100, 200\}$ and attributed it to high-HRS-tail exhaustion. The present paper retains the collapse statistic on the HP-driven trajectory but shows it disappears under HRS-blind drivers, revising the interpretation from "intrinsic property of hygiene-augmented scoring at high capacity" to "closed-loop artefact of recursive deployment".

EPSS dynamics. Jacobs et al. [8], [9] established per-CVE exploit-prediction scoring. Ravalico et al. [10] studied per-CVE EPSS score drift. The present paper documents a distinct effect: EPSS-as-policy remediation at high capacity exhausts the ground-truth positive set within six windows on the simulated fleet, producing a literal zero floor for every scorer evaluated against that trajectory. Neither prior study addressed the policy-level effect.

Counterfactual evaluation in security ML. The general pattern — evaluating method A on the data trajectory that method B would have generated — appears in counterfactual learning to rank and off-policy evaluation. The instance studied here is the simplest form: a deterministic policy generates the trajectory, and the counterfactual is computed by re-running the simulation under each candidate policy. We do not survey the broader counterfactual-policy literature; the question this paper resolves is operationally specific to the Paper 6 H4 claim.

Cyber-hygiene benchmarking. HygieneBench [11], HygienePrio [4], the temporal extension [2], capacity sweep [1], online calibration [3], and multi-history calibration [7] together constitute the methodological substrate within which this paper sits.

Policy mandates. CISA BOD 22-01 [12] and 23-01 [13] along with NIST SP 800-40 Rev. 4 [14] define the operational scheduling regime under which any vulnerability-prioritization deployment must operate. The recommendation that scoring and selection should be decoupled at high capacity follows from this paper's findings and aligns with the spirit of the BOD mandates' separation of discovery, prioritisation, and remediation roles.

XI. Conclusion

A pre-registered 7,500-row self-trajectory evaluation of the five HygienePrio-family prioritization methods at $K \in \{50, 200\}$, $\lambda = 3$, $W = 6$, on the 25 evaluation seeds inherited from Papers 5–8, produces three findings that retroactively reframe the research sequence.

First, Paper 6's headline "HygienePrio collapses at high capacity" is rejected as an intrinsic property of hygiene-augmented scoring. At $K = 200, w = 6$, HygienePrio-full's mean P@50 is 0.075 on its own trajectory but 0.701 on CVSS-driven, 0.713 on HRS-driven, and 0.706 on Random-driven trajectories. The collapse is a recursive-deployment artefact: HygienePrio's own selection preferentially drains the HRS upper tail that HygienePrio's score depends on.

Second, EPSS-only's selection at $K = 200$ produces a literal zero W6 P@50 for every scorer (range 0.001–0.012). The ground-truth positive set is structurally exhausted by EPSS-only's aggressive remediation of the high-EPSS tail. EPSS-self-driven at high capacity is the deepest sink in the studied grid.

Third, HygienePrio's per-pair dominance over EPSS-only is trajectory-conditional. At K=50 it is 1.000 under every driver; at K=200 it is 1.000 under HRS, CVSS, and Random drivers, 0.800 under HP's own driver, and 0.393 under EPSS-only's driver (where both methods collapse together). Paper 5's 96% headline statistic is preserved at moderate capacity and under HRS-blind drivers — the regimes most relevant to actual deployment.

The operational reading is sharper than Papers 5–8 allowed: HygienePrio is a strict improvement over EPSS-only across the studied regimes whenever the deploying organisation does not let HygienePrio also drive the patching queue at high capacity. The Paper 6 H4 collapse is a known closed-loop hazard, not a property of the scoring method itself, and is operationally controllable by decoupling scoring from selection.

Reproducibility. The runner, analysis script, pre-registration protocol, and frozen 7,500-row results CSV are available at https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper9.

Future work. (i) A pre-registered re-test of Paper 6's H4 hypothesis under the qualified statement "on every cell, on at least one non-HP driver". Doing this rigorously requires a fresh seed set to avoid HARKing on the present data. (ii) Trajectory effects on calibration: do Paper 7's lag-1 recovery ratio and Paper 8's smoothing penalty change under non-HP drivers? Both papers measured under T_HP only. (iii) External validation on real fleet telemetry with longitudinal exploitation outcome data. (iv) A cross-trajectory evaluation in which the scorer also predicts the next-window driver, modelling a fully adaptive closed-loop policy rather than two static ones.

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